



Effect of organic and inorganic fertigation on yields, $\delta^{15}\text{N}$ values, and $\delta^{13}\text{C}$ values of tomato (*Lycopersicon esculentum* Mill. cv. Saturn)

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Received 3 May 2002. Accepted in revised form 2 April 2003

Key words: $\delta^{13}\text{C}$, $\delta^{15}\text{N}$, fertigation, tomato, yield

Abstract

We examined the effects of fertilizer application, especially the effects of fertigation and types of fertilizer (inorganic and organic) on yields and $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ values of tomato (*Lycopersicon esculentum* Mill. cv. Saturn). Fertigation is a method in which an appropriate diluted liquid fertilizer is applied to the plants each time they are drip-irrigated. We developed a method of organic fertigation using corn steep liquor (CSL) as the liquid fertilizer, because it is an industrial byproduct of cornstarch manufacture and can be used very effectively. We compared fruit yield, mineral content, $\delta^{15}\text{N}$ value, and $\delta^{13}\text{C}$ value of tomatoes grown under three different fertilizer treatments, basal dressing: basal dressing with granular chemical fertilizer; inorganic fertigation: fertigation with liquid chemical fertilizer; and organic fertigation: fertigaion with CSL. Mineral contents of tomatoes grown with basal dressing were generally lower than those grown under either fertigation treatment. These results indicated that yields and mineral contents were influenced more by the method of fertilizer application than by whether the fertilizers were inorganic or organic. There were, however, significant differences in the $\delta^{15}\text{N}$ values of tomato fruits grown under different types of fertilizer applications, especially between inorganic and organic fertilizers. The $\delta^{15}\text{N}$ value of the chemical fertilizer used for basal dressing was $0.81 \pm 0.45\text{‰}$, that of the chemical fertilizer for fertigation was $0.00 \pm 0.04\text{‰}$, and that of CSL was $8.50 \pm 0.71\text{‰}$. The $\delta^{15}\text{N}$ values of the soils reflected the $\delta^{15}\text{N}$ values of the fertilizers. Moreover, the $\delta^{15}\text{N}$ values of the fruits corresponded to the $\delta^{15}\text{N}$ values of the applied fertilizers. The $\delta^{15}\text{N}$ values were $3.18 \pm 1.34\text{‰}$ in the fruits grown with a basal dressing of chemical fertilizer, $0.30 \pm 0.61\text{‰}$ in those grown under inorganic fertigation, and $7.09 \pm 0.68\text{‰}$ in those grown under organic fertigation. On the other hand, although the $\delta^{13}\text{C}$ values in the soil also reflected the $\delta^{13}\text{C}$ values of the applied fertilizers, there was no significant difference in the $\delta^{13}\text{C}$ values of fruits among the different treatments. In conclusion, because the $\delta^{15}\text{N}$ values of fertilizers correlated well with those of the fruits, it may be possible to use $\delta^{15}\text{N}$ values as an indicator of organic products.

Introduction

In current times, both in Japan and abroad, consumers are demanding higher quality and safer food, and they are becoming increasingly interested in organic products. However, researchers question the touted benefits of such organic food (Adams, 2001). Despite the ambiguity of the situation, public opinion has swayed

the government, and laws governing certification of organic products have been in effect in Japan since April 2001. In practice, however, the method used to certify organic products is an oral interview rather than an actual inspection, and identifying whether products have actually been cultivated using organic fertilizers remains difficult and ambiguous.

With this problem in mind, we firstly examined whether there was any difference in yields or mineral

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Table 1. Mineral contents of fertilizers

Fertilizer	N	P	K (g kg ⁻¹)	Ca	Mg
Ammonium Nitrate	334	—	—	—	—
Superphosphate	—	76	—	—	—
Potassium sulfate	—	—	415	—	—
Compound fertilizer S646	160	61	133	—	—
Ca Mg Fertilizer	—	—	—	—	—
OK-F-1	150	35	141	43	12
	33				
Corn Steep Liquor	(Ammonium-N, 3) (Nitrate-N, 0.4)	15	27	0.3	7
Oyster shell liming material	—	—	—	344	4

Guaranteed analysis of fertilizers.

—: not determined.

contents of fruit grown with organic or inorganic fertilizers under different practical application methods. Secondly, we investigated a method of numerically certifying organic products by using $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ values as indicators of organic products. We thought that active application of organic waste as fertilizer should be respected as environmentally sound, even if there were no tangible benefits in terms of yield or mineral content, as suggested by Smith (1993).

Previous studies have shown that different sources of N in the form of chemical fertilizers (basal and top dressing) and livestock manures applied to paddy fields differently affect the natural $\delta^{15}\text{N}$ in the rice plants (Tokunaga et al., 2000; Yoneyama et al., 1990). By measuring the ^{15}N signatures of various soils and plants, it is possible to determine the origins of the nitrogen pools (Yoneyama, 1996). These findings prompted us to investigate adopting $\delta^{15}\text{N}$ values as a marker for organic products. In a similar fashion, $\delta^{13}\text{C}$ values of soils were found to differ with different cropping management practices and were mainly affected by whether the cultivated plants were C_3 or C_4 (Balesdent et al., 1990). On the basis of these results, we expected that the $\delta^{15}\text{N}$ values of fruiting vegetables would reflect the different fertilizers that had been applied and the $\delta^{13}\text{C}$ values of the soil would reflect the application of plant residues.

In this paper, we measured the $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ values of tomatoes and soils after applying organic or inorganic fertilizer and estimated the correlation between the values of the fruits and soils. We discussed the possibility of using these values as an indicator of organic products.

Table 2. Total fertilizer application rate during each cultivation period

Treatment	Fertilizer	Fertilizer rate (g plant ⁻¹)		
		1999 EC*	1999 LC**	2000 EC
Basal dressing	Ammonium Nitrate	21.8	21.8	0
	Superphosphate	21.5	21.5	0
	Potassium sulfate	12	12	0
	Compound fertilizer S646	0	0	53.1
	Ca Mg Fertilizer	22.5	22.5	22.5
Inorganic fertigation	OK-F-1 (top dressing)	20	0	6.7
	OK-F-1	70	50	56.7
Organic fertigation	Corn Steep Liquor	317	227	257
	Oyster shell liming material	15	15	15

*EC: early cultivation, **LC: late cultivation.

Materials and methods

Plant materials and cultivation

Tomatoes (*Lycopersicon esculentum* Mill. cv. Saturn) were cultivated under three different fertilizer treatments: basal dressing (mainly used chemical granular fertilizer, Table 1), inorganic fertigation (inorganic liquid fertilizer), and organic fertigation (organic liquid fertilizer). We established an organic fertigation system by using corn steep liquor (CSL) as a liquid fertilizer instead of the usual inorganic fertilizer (Nakano et al., 2001). CSL is a byproduct of cornstarch manufacture and can be used very effectively. It is a viscous liquid and contains 33 g kg⁻¹ of N, 15 g kg⁻¹ of P and 27 g kg⁻¹ of K, and so on (Table 1). We measured Nitrate and Ammonium forms of N of CSL, which were 0.4 and 3g kg⁻¹ respectively. Other nitrogens may consist of organic N, such as amino acid, peptide, and protein form N. Plots of tomatoes were cultivated in isolated planters (30 cm depth × 83 cm width × 940 cm length) filled with Taketoyo sandy clay loam (total carbon: 0.61%, total nitrogen: 0.07%, CEC: 7.4 cmol (+) kg⁻¹) sieved through a 1 cm mesh and mixed with fertilizer described in Table 1. From May 1999 to August 2000, three successive cultivations were carried out at almost half-yearly intervals (Table 2). In each of these treatments, the total amounts of applied nitrogen were the same. Sixty 30-day-old tomato seedlings were planted in each plot without replication. For yield evaluation, we chose 15, 6, and 8 tomato plants from each plot arbitrarily in 1999EC, 1999LC, and 2000EC, respectively. For $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ evaluation, we chose 3, 3 and 4 samples of leaf, stem and fruits, respectively. For inorganic con-

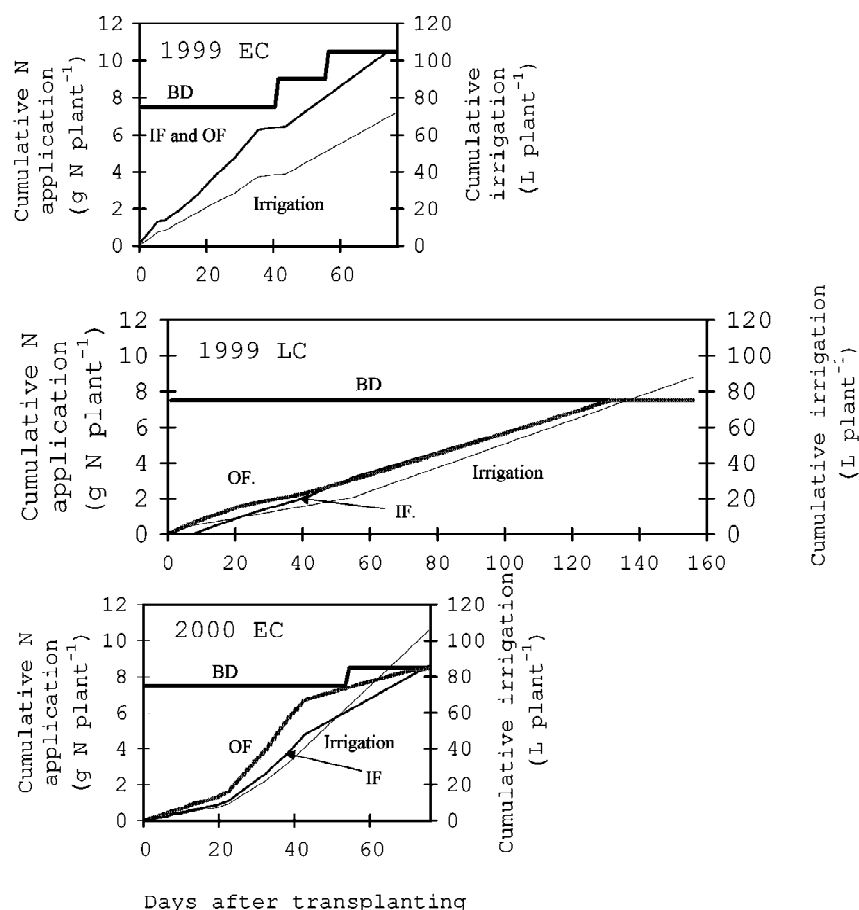


Figure 1. Cumulative nitrogen and water application rates in the 3 different types of fertilizer application. BD: Basal dressing, IF: Inorganic fertilization, OF: Organic fertilization, EC: early cultivation, LC: Late cultivation.

tent of soil and tomato plants, we chose 4 samples of them. A drip fertigation system (Netafim Inc. Israel) was installed for irrigation and fertilizer application. Drippers were set 10 cm from the base of each seedling. The first cultivation period was between 18 May and 2 August 1999 (1999 early cultivation: EC). The second cultivation was between 7 October 1999 and 10 March 2000 (1999 late cultivation LC). In the late cultivation, heaters were installed in the greenhouse set at 12 °C. The third cultivation was between 8 June and 22 August 2000 (2000 EC). The patterns of water and fertilizer applied during these periods are shown in Figure 1. In each of the three cultivations, tomato fruits were harvested and weighed once or twice every week after the fruits had turned red.

Measurement of $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ values and mineral contents We freeze-dried the fourth truss of tomatoes from the third cultivation and analyzed the mineral

contents, and $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ values. We calculated $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ using the following equation:

$$\delta^{15}\text{N} \text{ or } \delta^{13}\text{C} = (R_{\text{sample}} - R_{\text{standard}}) / R_{\text{standard}} \times 1000\text{‰},$$

where R is $^{15}\text{N}/^{14}\text{N}$ in the case of $\delta^{15}\text{N}$ and where R is $^{13}\text{C}/^{12}\text{C}$ in the case of $\delta^{13}\text{C}$ (Yoneyama and Sasakawa, 1994). At the same time, soil was also sampled from a distance of 10 cm from the drip point to a depth of 10 cm using a 5 cm-diameter core sampler for nutrient residue evaluation. The remaining upper parts of the tomato plants were vertically divided into four equal parts. Then the fertilizers, soil samples, and uppermost parts of the plants were oven-dried at 80 °C. The fertilizers, soil samples, upper parts of the plants, and freeze-dried fruits were each pulverized and sieved through a 0.5 mm mesh. The samples were enclosed in tin cups and analyzed by mass spectrometry (ANCA-SL, Europa Scientific, UK) to determine the $\delta^{15}\text{N}$

Table 3. Mineral contents of tomatoes grown under different fertilizer applications in 2000 EC cultivation

	N	P	K (g kg ⁻¹)	Ca	Mg
BD	16.1 ab	6.97 a	30.4 a	0.73 a	1.81 a
IF	14.3 a	9.93 b	50.0 b	1.33 b	2.61 b
OF	17.0 b	10.25 b	49.0 b	1.06 ab	2.53 b

BD: Basal dressing, IF: Inorganic fertigation, OF: Organic fertigation.

Values are means of 4 samples.

Means within the same column followed by the same letter are not significantly different at $p < 0.05$ (Fisher's LSD).

and $\delta^{13}\text{C}$ values. The samples were digested by concentrated nitric acid and the mineral contents were measured by ICP (Inductively-Coupled Plasma Spectrometry, SPS7700, Seiko Instrument, Japan).

Results and discussion

There was no significant difference between the yields of tomatoes grown under either the organic or the inorganic fertigation treatment (Figure 2). The yield of tomatoes grown under the basal dressing treatment was lower than under the fertigation treatments in the second cultivation (1999 LC) but higher in the third cultivation (2000 EC). The yield under basal dressing was lower in the second cultivation, because the air temperature was lower than in the earlier cultivation (1999 EC). The rate of nutrient release from the granular fertilizer applied as the basal dressing depends on soil temperature and, even though greenhouse temperatures remained above 12 °C throughout winter, the plants could more effectively absorb nutrients from fertigation. On the other hand, in the 2000 EC plots, basal dressing showed higher yields than the fertigation treatments, because in the fertigation treatments the fertilizer application during the first 30 days after transplanting was too low (Figure 1). The retardation of early growth later affected the yield. These results indicate that yields were more greatly affected by the method of fertilizer application (basal dressing or fertigation) than by the kind of fertilizer (organic or inorganic).

Table 3 shows the mineral content of tomatoes from the fourth truss from the 2000 EC plots. In this set of plantings, K, P, Ca, and Mg contents of tomatoes grown under fertigation treatments were greater than those grown under basal dressing. As shown in the fer-

Table 4. Mineral content in the soil after the cultivation of 2000 EC

	C	N	P (g kg ⁻¹)	K (g kg ⁻¹)	Ca	Mg
BD	26.3 a	1.76 a	6.90 b	5.01 a	19.9 b	4.9 b
IF	24.0 a	1.19 a	4.28 a	3.75 a	9.4 a	3.0 a
OF	43.2 b	4.29 b	6.88 b	4.78 a	6.4 a	4.9 b

BD: Basal dressing, IF: Inorganic fertigation, OF: Organic fertigation.

Values are means of 4 samples.

Means within the same column followed by the same letter are not significantly different at $p < 0.05$ (Fisher's LSD).

tilizer application pattern (Figure 1), applied fertilizers were abundant during the latter period of cultivation under the fertigation treatments. On the other hand, because nitrogen is easily transferred, there was no significant difference between the nitrogen content of BD and that of IF or OF. But the nitrogen content of OF is higher than that of IF, because applied organic N was released slower than in IF treatment (Nakano et al., 2001). Smith (1993) showed that crops grown with organic fertilizers had higher mineral contents than conventionally cultivated crops. However, our experiment showed that the mineral content of tomatoes as well as fruit yield was affected more by the way that the fertilizer was applied than by the type of fertilizer (inorganic or organic). By adopting a unique method of applying liquid organic fertilizer to use the organic byproducts effectively instead of applying conventional organic manure (for example, cattle manure), we were able, to some extent, to compare more precisely how inorganic and organic fertilizer applications affected the plants.

Table 4 shows mineral concentrations in the soil after cultivation. Because the soil samples were collected 10 cm away from the dripping points, the concentrations measured did not always reflect those throughout the topsoil. In the organic fertigation, more carbon and nitrogen remained in the soil than in the other chemical fertilizer applications, suggesting that some of the applied organic material were stored in the soil. Although calcium is known as a relatively immobile element, its uptake was especially enhanced in the fertigations. The amount of calcium remaining in the soil was lower in the fertigations than in the conventional basal fertilizer application, because the fertigation could constantly apply moderate concentrations of calcium to the rhizosphere.

Yields and mineral contents were influenced more by the method of fertilizer application than by whether

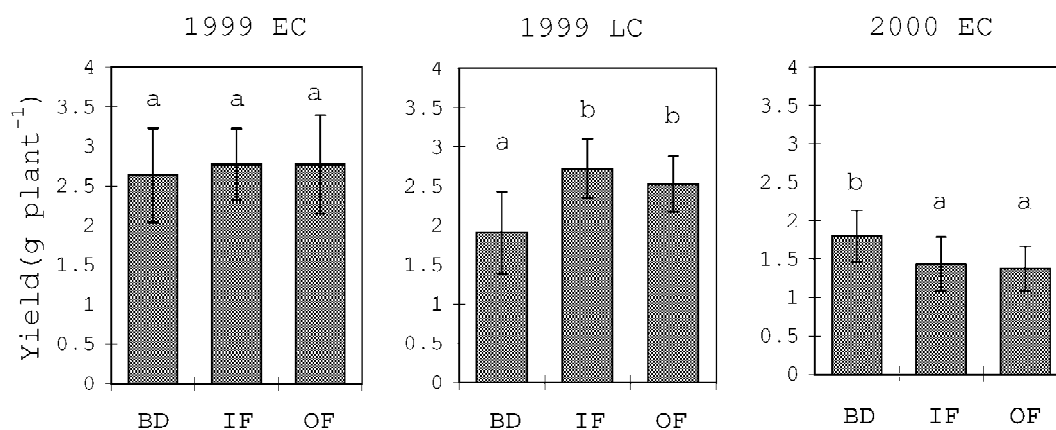


Figure 2. Yields of tomatoes grown under different fertilizer application regimes. BD: Basal dressing IF: Inorganic fertigation OF: Organic fertigation. Values for 1999 EC, 1999 LC, and 2000 EC are means of 15, 6, and 8 samples, respectively. Means with the same letter within each group are not significantly different at $p < 0.05$ (Fisher's LSD). Vertical bars indicate SD.

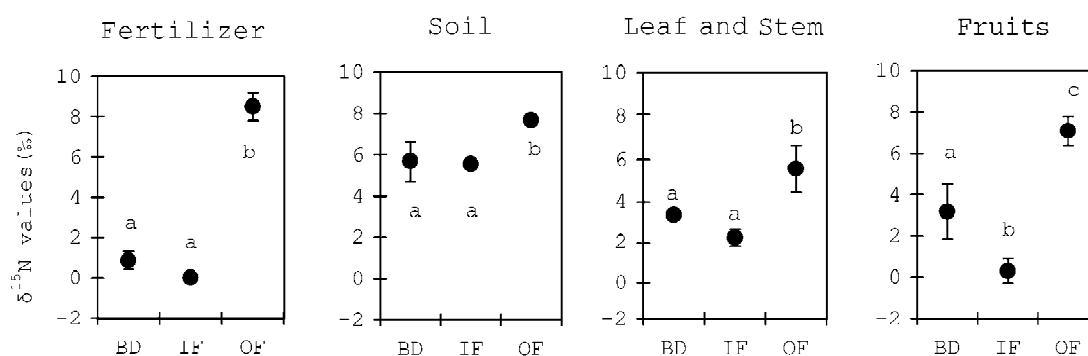


Figure 3. $\delta^{15}\text{N}$ values of fertilizer, mixtures of fertilizers and manure, soil, and fruits of tomato. BD: Basal dressing, IF: Inorganic fertigation, OF: Organic fertigation. Values of fertilizer, soil, leaf and stem, and fruit are means of 2, 3, 3, and 4 samples, respectively. Means within each group of fertilizer, soil, and fruit identified by the same letter are not significantly different at $p < 0.05$ (Fisher's LSD). Vertical bars indicate SD.

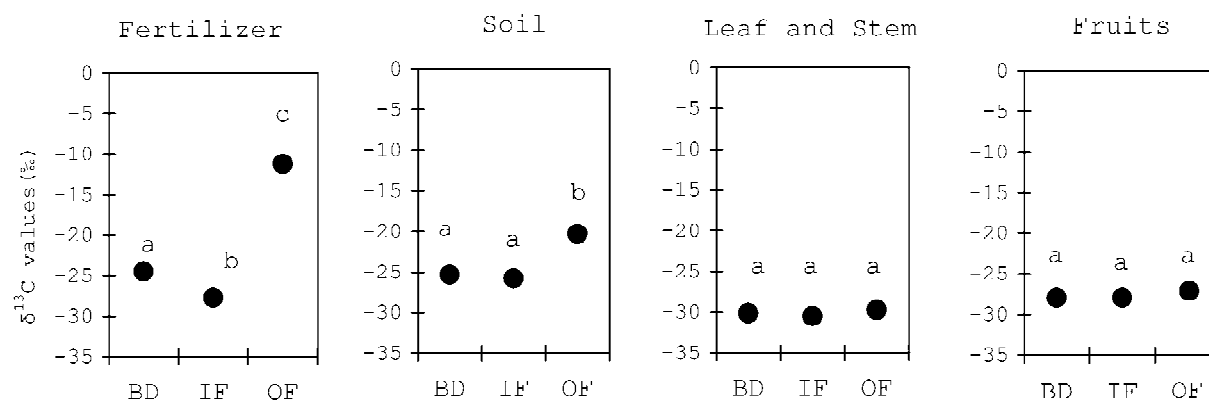


Figure 4. $\delta^{13}\text{C}$ value of fertilizer, soil, stem and leaf, and fruits of tomato. BD: Basal dressing, IF: Inorganic fertigation, OF: Organic fertigation. Values of fertilizer, soil, leaf and stem, and fruit are means of 2, 3, 3, and 4 samples, respectively. Means within each group of fertilizer, soil, and fruit identified by the same letter are not significantly different at $p < 0.05$ (Fisher's LSD). Vertical SD bars were included in the symbols.

the fertilizers were inorganic or organic. However, the $\delta^{15}\text{N}$ values differed among treatments. In particular, the $\delta^{15}\text{N}$ values of tomato fruits reflected the value of applied fertilizers (Figure 3). The $\delta^{15}\text{N}$ value of the chemical fertilizer for basal dressing was $0.81 \pm 0.45\text{‰}$, that of the chemical fertilizer for fertigation was $0.00 \pm 0.04\text{‰}$, and that of CSL was $8.50 \pm 0.71\text{‰}$. The $\delta^{15}\text{N}$ values of the soil reflected the $\delta^{15}\text{N}$ values of the fertilizers; moreover, the $\delta^{15}\text{N}$ value of the fruit corresponded to that of the applied fertilizer. The $\delta^{15}\text{N}$ value of the fruit was $3.18 \pm 1.34\text{‰}$ when grown under basal dressing, $0.30 \pm 0.61\text{‰}$ under inorganic fertigation, and $7.09 \pm 0.68\text{‰}$ under organic fertigation. Fertilizer applied as a basal dressing might release ammonia through vaporization (Yoneyama et al., 2001), and the more NO_3^- is applied, the more N_2O is emitted (Inubushi et al., 1999). Thus, basal dressing showed a higher ^{15}N discrimination in the chemical fertilizer treatments probably due to ammonia vaporization and denitrification, i.e. processes that are recognized as major steps in possible isotopic fractionation (Mariotti et al., 1977). In the BD, roots absorbed soil fertility nitrogen which showed higher than the $\delta^{15}\text{N}$ values of chemical fertilizer (Yoneyama, 1996). Because fertigation is a method whereby small quantities of nutrients are applied each time when the plants are irrigated, fertilizer use is expected to be more efficient (Hartz and Hochmuth, 1996; Hochmuth, 1992). Therefore, ammonia vaporization and denitrification, which contribute to isotope discrimination, rarely happen and tomatoes do not depend on nitrogen for soil fertility in the fertigation. $\delta^{15}\text{N}$ values of tomato fruits cultivated with fertigation using chemical fertilizer were lower than those of basal dressing because tomato plants took up the nitrogen before the effects of the isotope discrimination processes began.

The $\delta^{15}\text{N}$ values of tomato fruits grown under the chemical fertilizer treatments were lower than those grown with organic fertilizer. These results indicated that $\delta^{15}\text{N}$ values might be an indicator for organic products. $\delta^{15}\text{N}$ values were higher in livestock manures than in chemical fertilizers (Tokunaga et al., 2000; Yamada et al., 1999); CSL also showed higher $\delta^{15}\text{N}$ values than chemical fertilizer because of the isotope discrimination occurring during the growth of the maize and the CSL production process. The $\delta^{15}\text{N}$ value of the CSL corresponded well to those of the tomato fruits. This result indicated that the nitrogen derived from the organic liquid fertilizer was absorbed by the plants before the isotope discrimination process happened in the soil. Thus, the most efficient use of

nutrients might be achieved not only through inorganic fertigation but also through organic fertigation.

As for $\delta^{13}\text{C}$ values, because CSL is derived from maize, which is a C_4 plant, the CSL itself showed higher $\delta^{13}\text{C}$ values than the chemical fertilizers (Figure 4). However, although the $\delta^{13}\text{C}$ values of the soil reflected the $\delta^{13}\text{C}$ values of the fertilizers, the $\delta^{13}\text{C}$ values of the fruits did not show any difference among the fertilizer treatments. These results suggested that almost all of the carbon found in the fruits was derived from atmospheric carbon, and not from the soil or fertilizer.

There is some possibility to use $\delta^{15}\text{N}$ values as an indicator of organic products because the $\delta^{15}\text{N}$ values found in fruits correlated well with the $\delta^{15}\text{N}$ values of the applied fertilizers. The boundary which distinguishes ordinal from organic products would be around 5‰ . However, in organic agriculture (compost which showed high $\delta^{15}\text{N}$ values (Tokunaga et al., 2000)) mainly used in Japan, there may be some exceptions, for example, management urine which showed low $\delta^{15}\text{N}$ values as a chemical fertilizer (Steele and Daniel, 1977) was used. Therefore, further studies using other soils and other types of organic fertilizers are needed to use $\delta^{15}\text{N}$ values for certification of organic products.

Acknowledgements

We thank Dr Y. Ohwaki and his staff of the experimental research stations (National Agricultural Research Center) who kindly supported the analysis of $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ values.

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